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CONSTANT POWER ACTIVE DISCHARGE OF ENERGY STORAGE CIRCUIT

Abstract

In some examples, a circuit includes a reference signal generator, a comparator, a one-shot circuit, and a gate driver. The reference signal generator has first and second input terminals and an output terminal, the first input terminal of the reference signal generator configured to receive a discharge time select value, and the second input terminal of the reference signal generator configured to receive a stop voltage select value. The comparator has first and second input terminals, and an output terminal, the first input terminal of the comparator configured to receive a sensed voltage, and the second input terminal of the comparator coupled to the output terminal of the reference signal generator. The one-shot circuit has first and second input terminals, and an output terminal, the first input terminal coupled to the output terminal of the comparator, and the second input terminal configured to receive an on-time value. The gate driver has an input terminal and an output terminal, the input terminal of the gate driver coupled to the output terminal of the one-shot circuit.

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Background/Summary

BACKGROUND

[0001] Various electrically powered systems may include energy storage circuits, such as capacitors. In some application environments, these energy storage circuits may pose a risk to users, such as if they retain a charge under certain operational circumstances, or retain a charge for greater than a threshold amount of time under certain operational circumstances.

SUMMARY

[0002] In some examples, a circuit includes a reference signal generator, a comparator, a one-shot circuit, and a gate driver. The reference signal generator has first and second input terminals and an output terminal, the first input terminal of the reference signal generator configured to receive a discharge time select value, and the second input terminal of the reference signal generator configured to receive a stop voltage select value. The comparator has first and second input terminals, and an output terminal, the first input terminal of the comparator configured to receive a sensed voltage, and the second input terminal of the comparator coupled to the output terminal of the reference signal generator. The one-shot circuit has first and second input terminals, and an output terminal, the first input terminal coupled to the output terminal of the comparator, and the second input terminal configured to receive an on-time value. The gate driver has an input terminal and an output terminal, the input terminal of the gate driver coupled to the output terminal of the one-shot circuit.

[0003] In some examples, a circuit is configured to receive a sensed voltage representative of a voltage of an energy storage circuit. The circuit is also configured to compare the sensed voltage to a reference signal representative of a discharge profile for the energy storage circuit. The circuit is also configured to, responsive to the sensed voltage having a value greater than the reference signal, control a gate driver to provide a control signal configured to discharge the energy storage circuit for a constant on-time period determined to cause a power discharge rate of the energy storage circuit to remain constant.

[0004] In some examples, a system includes a switch, a capacitor, a voltage sense circuit, a high-side drive circuit, a first transistor, a low-side drive circuit, and a second transistor. The switch has first and second terminals, the second terminal of the switch coupled to a positive power terminal. The capacitor has first and second terminals, the first terminal of the capacitor coupled to the first terminal of the switch, and the second terminal of the capacitor coupled a ground terminal. The voltage sense circuit has first and second terminals and an output terminal, the first terminal of the voltage sense circuit coupled to the first terminal of the switch, and the second terminal of the voltage sense circuit coupled to the ground terminal. The high-side drive circuit has an input terminal and an output terminal. The first transistor has first and second terminals, and a control terminal, the first terminal of the first transistor coupled to the first terminal of the switch, and the control terminal of the first transistor coupled to the output terminal of the high-side drive circuit. The low-side drive circuit has first and second input terminals and an output terminal, the first input terminal of the low-side drive circuit coupled to the output terminal of the voltage sense circuit. The second transistor has first and second terminals, and a control terminal, the first terminal of the second transistor coupled to the second terminal of the first transistor, the second terminal of the second transistor coupled to the ground terminal, and the control terminal of the second transistor coupled to the output terminal of the low-side drive circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** is a block diagram of a system, in accordance with various examples.
[0006] FIG. **2** is a block diagram of a drive circuit, in accordance with various examples.
[0007] FIG. **3** is a diagram of signal waveforms, in accordance with various examples.
[0008] FIG. **4** is a diagram of signal waveforms, in accordance with various examples.
[0009] FIG. **5** is a diagram of signal waveforms, in accordance with various examples.
[0010] FIG. **6** is a flowchart of a method, in accordance with various examples.
DETAILED DESCRIPTION
[0011] As described above, various electrically powered systems may include energy storage circuits, such as capacitors. An example of such an electrically powered system may be an electrically powered vehicle, a hybrid power vehicle, an electric subsystem of a vehicle, or the like. As used herein, vehicle may include automobiles, trucks, boats, airplanes, helicopters, industrial (e.g., construction, farm, or the like) equipment, or any other suitable wheeled or tracked terrestrial vehicle, marine vehicle, or airborne vehicle. In some application environments, these energy storage circuits may pose a risk to users, such as if they retain a charge under certain operational circumstances, or retain a charge for greater than a threshold amount of time under certain operational circumstances. For example, a vehicle traction inverter system of a battery-powered, or hybrid-powered, vehicle may include a direct current (DC) link capacitor. The DC link capacitor may increase stability of a DC voltage provided to load, such as a vehicle traction inverter system, by supplementing power provided by a battery or other power source in response to sporadic or transient heavy-load conditions imposed by the vehicle traction inverter system on the power source. Even after the DC link capacitor is decoupled from a power source (e.g., battery) of the vehicle, the DC link capacitor may retain a charge and resulting voltage. In some examples, the voltage may be in a range of 800 volts (V) or greater. The DC link capacitor voltage may pose a risk to users in certain operational circumstances, such as in the event of a crash of the vehicle, during maintenance or other service of the vehicle, or the like. As a result, some safety regulations, regulations, laws, standards, or industry practices may specify a time duration and/or discharge rate for the DC link capacitor. For example, the time duration and/or discharge rate may specify that a voltage of the DC link capacitor should be decreased (e.g., discharged) to less than a threshold voltage within a permissible time range. In an example, the time range may be about 1 to about 5 seconds. In some examples, the threshold voltage may be about 60 V, about 40 V, between about 60 V and about 40 V, or any other suitable level or range of levels. Some approaches to discharging the DC link capacitor include discharging stored energy through a resistor. However, such approaches may generate large amounts of heat, may require comparatively large resistors (e.g., in physical size and/or resistance), or may otherwise be undesirable or unsuitable for certain application environments.
[0012] Examples of this description provide for active discharge of an energy storage circuit. In an example, the energy storage circuit is a capacitor, such as a DC link capacitor. In some examples, the active discharge provides for a constant power discharge rate in which a small change in voltage over time is seen at high voltages and a large change in voltage over time is seen at low voltages. The constant power discharge rate may be in contrast to a constant current discharge rate, reducing a peak junction temperature of power switches (e.g., transistors) in a critical path of the discharge. In an example, a voltage sensing circuit may provide a sensed signal indicative or otherwise representative of a DC link voltage (e.g., a voltage of the DC link capacitor). The voltage sensing circuit may be, in some examples, a voltage divider coupled in parallel with the DC link capacitor. The sensed signal may be compared to a reference signal. Based on that comparison, the DC link capacitor may be discharged. For example, the reference signal may be a reference

waveform representative of a discharge waveform for the DC link capacitor. Responsive to the sensed signal having a value greater than the reference signal, and a clock signal having an asserted value, a gate driver drives a switch (e.g., a transistor) to discharge the DC link capacitor. In some examples, the gate driver drives the switch based on a pulsed input signal that de-asserts after a programmed amount of time. In this way, control is exerted over a rate of discharge of the DC link capacitor. By controlling the rate of discharge of the DC link capacitor, an increased amount of control may be provided over thermal characteristics of the system (e.g., power switches in a critical path of the discharge, passive components, etc.), increasing performance of the system, rendering the system suitable for implementation in a greater range of application environments, or the like. In some examples, to discharge the DC link capacitor, the switch (or multiple switches) are configured to create a path to ground (e.g., a short to ground) between the DC link capacitor and a ground terminal at which a ground voltage potential is provided. The switch (or switches) may be controlled to create the path to ground based on control exerted by the gate driver.
[0013] By modulating a control voltage (e.g., gate voltage) of the switches, the switches are controlled to create multiple short circuit events, discharging the DC link capacitor in a controlled manner. The control voltage is modulated based on a comparison of the sensed signal to the reference signal to implement pulse-frequency modulation (PFM) control. For example, a frequency of the short circuit events increases as the DC link voltage decreases.
[0014] FIG. **1** is a block diagram of a system **100**, in accordance with various examples. In an example, the system **100** includes a power source **102** (e.g., a battery, an array of batteries, a battery cell, etc.), a switch **103**, a high-side drive circuit **104**, a transistor **106**, a low-side drive circuit **108**, a transistor **110**, a high-side drive circuit **112**, a transistor **114**, a low-side drive circuit **116**, a transistor **118**, a high-side drive circuit **120**, a transistor **122**, a low-side drive circuit **124**, a transistor **128**, a capacitor **130**, a voltage sense circuit **131**, and a motor **132**. In some examples, the capacitor **130** is a DC link capacitor. In an example, the motor **132** is a three-phase motor, having Phase A, Phase B, and Phase C terminals. The high-side drive circuit **104**, transistor **106**, low-side drive circuit **108**, and transistor **110** may form a first branch for driving a first phase of the motor **132**, the high-side drive circuit **112**, transistor **114**, low-side drive circuit **116**, and transistor **118** may form a second branch for driving a second phase of the motor **132**, and the high-side drive circuit **120**, transistor **122**, low-side drive circuit **124**, and transistor **128** may form a third branch for driving a third phase of the motor **132**. In some examples, the system **100** also includes a controller **134**. In an example, the controller **134** controls operation of other components of the system **100**, such as the high-side drive circuits **104**, **112**, **120**, the low-side drive circuits **108**, **116**, **124**, and/or the switch **103**.
[0015] In an example architecture of the system **100**, the power source **102** has a positive terminal and a negative, or ground, terminal. The switch **103** has first and second terminals, the second terminal of the switch **103** coupled to the positive terminal of the power source **102**. The capacitor **130** has first and second terminals, the first terminal of the capacitor **130** coupled to the first terminal of the switch **103**, and the second terminal of the capacitor **130** coupled to the ground terminal of the power source **102**. The voltage sense circuit **131** has an output terminal and first and second terminals, the first terminal of the voltage sense circuit **131** coupled to the first terminal of the switch **103**, and the second terminal of the voltage sense circuit **131** coupled to the ground terminal of the power source **102**. The high-side drive circuit **104** has a first input terminal at which a low-voltage power supply (LV_BAT), which may be derived from the power source **102** or from any other suitable source, is provided, a second input terminal coupled to the controller **134**, a third input terminal, and an output terminal. The transistor **106** has a first terminal coupled to the first terminal of the switch **103**, a second terminal coupled to the third input terminal of the high-side drive circuit **104**, and a control terminal coupled to the output terminal of the high-side drive circuit **104**. The low-side drive circuit **108** has a first input terminal coupled to the positive terminal of the power source **102**, a second input terminal coupled to the controller **134**, a third input terminal

coupled to the ground terminal of the power source **102**, a fourth input terminal coupled to the output terminal of the voltage sense circuit **131**, and an output terminal. The transistor **110** has a first terminal coupled to the second terminal of the transistor **106**, a second terminal coupled to the ground terminal of the power source **102**, and a control terminal coupled to the output terminal of the low-side drive circuit **108**. In an example, the second terminal of the transistor **106** is coupled to the Phase A terminal of the motor **132**.

[0016] The high-side drive circuit **112** has a first input terminal at which LV_BAT is provided, a second input terminal coupled to the controller **134**, a third input terminal, and an output terminal. The transistor **114** has a first terminal coupled to the first terminal of the switch **103**, a second terminal coupled to the third input terminal of the high-side drive circuit **112**, and a control terminal coupled to the output terminal of the high-side drive circuit **112**. The low-side drive circuit **116** has a first input terminal coupled to the positive terminal of the power source **102**, a second input terminal coupled to the controller **134**, a third input terminal coupled to the ground terminal of the power source **102**, a fourth input terminal coupled to the output terminal of the voltage sense circuit **131**, and an output terminal. The transistor **118** has a first terminal coupled to the second terminal of the transistor **114**, a second terminal coupled to the ground terminal of the power source **102**, and a control terminal coupled to the output terminal of the low-side drive circuit **116**. In an example, the second terminal of the transistor **114** is coupled to the Phase B terminal of the motor **132**.

[0017] The high-side drive circuit **120** has a first input terminal at which LV_BAT is provided, a second input terminal coupled to the controller **134**, a third input terminal, and an output terminal. The transistor **122** has a first terminal coupled to the first terminal of the switch **103**, a second terminal coupled to the third input terminal of the high-side drive circuit **120**, and a control terminal coupled to the output terminal of the high-side drive circuit **120**. The low-side drive circuit **124** has a first input terminal coupled to the positive terminal of the power source **102**, a second input terminal coupled to the controller **134**, a third input terminal coupled to the ground terminal of the power source **102**, a fourth input terminal coupled to the output terminal of the voltage sense circuit **131**, and an output terminal. The transistor **126** has a first terminal coupled to the second terminal of the transistor **122**, a second terminal coupled to the ground terminal of the power source **102**, and a control terminal coupled to the output terminal of the low-side drive circuit **124**. In an example, the second terminal of the transistor **122** is coupled to the Phase C terminal of the motor **132**.

[0018] The first branch, second branch, and third branch, as described above, are coupled in parallel with the capacitor **130** and each may be controlled to discharge the capacitor **130** individually, or in combination with another branch. For clarity of description, operation of discharging the capacitor **130** in the system **100** will be described with respect to the first branch, comprising the high-side drive circuit **104**, transistor **106**, low-side drive circuit **108**, and transistor **110**. However, in other examples, the second branch, or third branch, may alternatively, or additionally, be controlled to discharge the capacitor **130**, and description herein with respect to the first branch discharging the capacitor **130** may be equally applicable to the second and/or third branches.

[0019] In an example of operation of the system **100**, the controller **134** controls the high-side drive circuits **104**, **112**, **120**, the low-side drive circuits **108**, **116**, **124**, and/or the switch **103** to provide power from the power source **102** to the motor **132** to control or otherwise drive the motor **132**. In an example, the control may be according to a pulse-width modulation signal received from any suitable source, such as the controller **134**, the scope of which is not limited herein. As such, control of the high-side drive circuits **104**, **112**, **120**, the low-side drive circuits **108**, **116**, **124**, and/or the switch **103** to provide power to the motor **132** may be outside the scope of this disclosure, and description of which is omitted herein.

[0020] Responsive to opening of the switch **103** (e.g., de-coupling the high-side drive circuits **104**,

112, **120** and the low-side drive circuits **108**, **116**, **124** from the positive terminal of the power source **102**), the controller **134** may control one or more of the high-side drive circuits **104**, **112**, **120** and the low-side drive circuits **108**, **116**, **124** to discharge the capacitor **130**. The discharging may be performed according to PFM such that the capacitor **130** is discharged at an approximately constant power rate, with a change in voltage over time of a voltage of the capacitor increasing as the voltage of the capacitor decreases.

[0021] For example, responsive to assertion of a safe state enable signal by the controller **134**, the switch **103** may be opened. The safe state enable signal may be asserted by the controller **134** responsive to any suitable event, such as a key-off or turn-off event of the system **100**, the system **100** entering a maintenance or service mode, detection of a crash of the system **100**, detection of a fault or other anomalous condition of the system **100**, or the like. Also responsive to assertion of the safe state enable signal, the high-side drive circuit **104** may control the transistor **106** to remain in a conductive, or turned-on, state.

[0022] Responsive to opening of the switch **103**, the controller **134** may assert an active discharge enable signal. In some examples, asserting the active discharge enable signal includes providing the active discharge enable signal having a value of logic **1** (e.g., an active high signal). In other examples, asserting the active discharge enable signal includes providing the active discharge enable signal having a value of logic **0** (e.g., an active low signal). Responsive to assertion of the active discharge enable signal, the low-side drive circuit **108** may modulate a control signal provided at the control terminal of the transistor **110**. By modulating the control voltage, the low-side drive circuit **108** controls the transistor **110** to be in alternating conductive and non-conductive states, creating short circuit events between the first terminal of the capacitor **130** and the ground terminal of the power source **102**. During these short circuit events while the transistor **110** is conductive, a portion of charge stored by the capacitor **130** discharges, reducing a voltage of the capacitor **130**. In an example, each conductive period of the transistor **110**, and therefore each short circuit event, is a constant on-time event. In this way, each discharge period may have approximately a same duration. However, a frequency of the short circuit events, and therefore a frequency of the discharge periods, may increase as the voltage of the capacitor **130** decreases. In this way, the low-side drive circuit **108** modulates the control signal provided at the control terminal of the transistor **110** according to PFM.

[0023] In some examples, the low-side drive circuit **108** modulates the control signal based on a comparison of a sensed voltage (VSENSE) provided by the voltage sense circuit **131** and a reference voltage. In other examples, the low-side drive circuit **108** modulates the control signal based on a discharge control signal received from any suitable source, such as the controller **134**, the scope of which is not limited herein. In some examples, the voltage sense circuit **131** includes, or is implemented as, a resistor-based voltage divider (or voltage divider of any other suitable architecture) such that the sensed voltage is proportional to the voltage of the capacitor **130**. The low-side drive circuit **108** may compare the sensed voltage to a reference voltage or reference signal. In some examples, the reference signal is a waveform provided by a signal generator. The signal generator may generate the waveform based on a discharge time and a stop voltage, which may be received by the low-side drive circuit **108** from the controller **134**, received by the low-side drive circuit **108** as externally provided user input, retrieved by the low-side drive circuit **108** from a register or other storage device or location, or hard-coded to the low-side drive circuit **108**. Responsive to the sensed voltage having a value greater than or equal to the reference voltage, the low-side drive circuit **108** may provide the control signal to the transistor **110** having a pulse asserted for a programmed amount of time (e.g., an “on time”). The pulse may be an active-high pulse or an active-low pulse, as described above, depending on a process technology of the transistor **110**.

[0024] In some examples, the programmed amount of time may be received by the low-side drive circuit **108** from the controller **134**, received by the low-side drive circuit **108** as externally provided user input, retrieved by the low-side drive circuit **108** from a register or other storage device or location, or hard-coded to the low-side drive circuit **108**. In some examples, the low-side drive circuit **108** may limit a frequency of the modulation of the control signal. In such examples, the low-side drive circuit **108** may provide the control signal to the transistor **110** having the pulse asserted for the programmed amount of time responsive to both the sensed voltage having a value greater than or equal to the reference voltage, and a clock signal having an asserted value. In some examples, the clock signal may be provided by a clock generator based on the sensed voltage and a clock frequency limit value. The clock frequency limit value may be received by the low-side drive circuit **108** from the controller **134**, received by the low-side drive circuit **108** as externally provided user input, retrieved by the low-side drive circuit **108** from a register or other storage device or location, or hard-coded to the low-side drive circuit **108**. In some examples, the average discharge current is proportional to the clock signal. At high voltage, a lower frequency the clock signal is provided to maintain constant power discharge, as described herein. To mitigate excessive average discharge current during a fault scenario, the frequency of the clock signal may be scaled inversely proportional to the sensed voltage.

[0025] In some examples, the low-side drive circuit **108** may monitor the discharge process for the presence of a fault. For example, the low-side drive circuit **108** may compare the sensed voltage to a fault reference voltage comprising the reference voltage plus a fault offset value. The fault offset value may be received by the low-side drive circuit **108** from the controller **134**, received by the low-side drive circuit **108** as externally provided user input, retrieved by the low-side drive circuit **108** from a register or other storage device or location, or hard-coded to the low-side drive circuit **108**. Responsive to the sensed voltage exceeding the fault reference value, the low-side drive circuit **108** may assert a fault signal. Responsive to assertion of the fault signal, the low-side drive circuit **108** may disable discharge of the capacitor **130** by the low-side drive circuit **108**. The disabling of discharge of the capacitor **130** by the low-side drive circuit **108** responsive to assertion of the fault signal may be performed according to any suitable process, and via any additional supporting components (not shown), the scope of which is not limited herein. In some examples, the low-side drive circuit **108** may provide the fault signal to the controller **134** for communication to any one or more of the high-side drive circuits **104**, **112**, **120**, the low-side drive circuits **116**, **124**, and/or any other suitable devices. In other examples, the low-side drive circuit **108** may itself provide the fault signal to any one or more of the high-side drive circuits **104**, **112**, **120**, the low-side drive circuits **116**, **124**, and/or any other suitable devices.

[0026] In some examples, responsive to the low-side drive circuit **108** determining and reporting the fault, and consequently disabling discharge of the capacitor **130** by the low-side drive circuit **108**, one or more of the low-side drive circuit **116**, **124** may attempt to discharge the capacitor **130** in a manner substantially similar to that described above with respect to the low-side drive circuit **108**. In some examples, responsive to the low-side drive circuit **108** determining and reporting the fault, a notification may be provided to a user, or to another circuit, component, or device. The notification may take any form, such as a visual notification, an audible notification, a text notification, a notification transmitted via a network interface, an analog or digital signal transmitted between electrical components, or the like.

[0027] FIG. **2** is a schematic diagram of a drive circuit **200**, in accordance with various examples. In various examples, the drive circuit **200** is suitable for implementation as any one or more of the low-side drive circuits **108**, **116**, **124**. In an example, the drive circuit **200** includes a gate driver **202**, a reference signal generator **204**, a comparator **205**, a clock generator **206**, an AND logic circuit **208**, a one-shot circuit **210**, and a comparator **212**. The reference signal generator **204** and the clock generator **206** may each be implemented according to any suitable architecture, the scope of which is not limited herein. In some examples, the reference signal generator **204** is implemented as a lookup table which may provide a time to spend at each voltage level. For example, across a time range t**0** to t**1**, the reference signal generator **204** provides a voltage having a decreasing value beginning at V**1** and decreasing to V**1** in multiple discrete voltage levels as time increases from t**0** to t**1**. In some examples, the clock generator **206** is implemented using a combination of a lookup table that provides an increasing pulse-width modulation (PWM) frequency as voltage decreases, and a PWM generator that generates a PWM signal at the PWM frequency for each voltage level. In some examples, the one-shot circuit **210** may be implemented as a multivibrator, such as a monostable multivibrator. In other examples, the one-shot circuit **210** may be implemented according to any suitable architecture or as any suitable circuit capable of generating or providing a constant on time pulse signal responsive to a received signal. The gate driver **202** may be implemented according to any suitable architecture, the scope of which is not limited herein.

[0028] In an example, the reference signal generator **204** has an output terminal, and has a first input terminal configured to receive a safe state enable signal, a second input terminal configured to receive an active discharge enable signal, a third input terminal configured to receive a discharge time select value, and a fourth input terminal configured to receive a stop voltage select value. In various examples, any one or more of the first through fourth input terminals of the reference signal generator **204** may be coupled to respective output terminals of the controller **134**, or any other suitable signal sources, as described elsewhere herein. The comparator **205** has a first input terminal configured to receive a sensed voltage, a second input terminal coupled to the output terminal of the reference signal generator **204**, and has an output terminal. In some examples, the first input terminal of the comparator **205** is coupled to the output terminal of the voltage sense circuit **131** to receive the sensed voltage from the voltage sense circuit **131**. In some examples, the drive circuit **200** does not implement a limit on a frequency according to which the drive circuit implemented PFM control of the transistor **110**. In such examples, the output terminal of the comparator **205** may be coupled to an input terminal of the one-shot circuit **210**.

[0029] In other examples, the clock generator **206** has a first input terminal configured to receive the sensed voltage (e.g., coupled to the output terminal of the voltage sense circuit **131** to receive the sensed voltage from the voltage sense circuit **131**), a second input terminal configured to receive a clock frequency limit value, and has an output terminal. In some examples, the second input terminal of the clock generator **206** may be coupled to an output terminal of the controller **134**, or any other suitable signal source, as described elsewhere herein. The AND logic circuit **208** has a first input terminal coupled to the output terminal of the comparator **205**, a second input terminal coupled to the output terminal of the clock generator **206**, and has an output terminal. The one-shot circuit **210** has a first input terminal coupled to the output terminal of the AND logic circuit **208**, a second input terminal configured to receive an on-time value, and has an output terminal coupled to the gate driver **202**. In some examples, the second input terminal of the one-shot circuit **210** may be coupled to an output terminal of the controller **134**, or any other suitable signal source, as described elsewhere herein.

[0030] Operation of the drive circuit **200** will be described herein under an operational circumstance in which both the safe state enable signal and the active discharge enable signal have asserted values, as described above herein. In such an example of operation of the drive circuit **200**, the reference signal generator **204** provides a reference signal based on the discharge time select value (e.g., specifying t**1**) and stop voltage select value (e.g., specifying V**1**), which may specify a discharge time for discharging the capacitor **130** and a voltage to which the capacitor **130** is discharged in that discharge time. The comparator **205** receives the sensed voltage, which may be representative of a voltage of the capacitor **130**, from the voltage sense circuit **131**, and receives the reference signal from the reference signal generator **204**. In an example, the reference signal emulates the voltage waveform of a capacitor discharged at constant power. Responsive to the sensed voltage having a value greater than the reference signal, the comparator **205** provides a comparison result having a value of logic **1**. Responsive to the sensed voltage not having a value greater than the reference signal, the comparator **205** provides the comparison result have a value of logic **0**.

[0031] The clock generator **206** provides a clock signal (CLOCK) based on the clock frequency limit value, which may specify a maximum frequency of a control signal for controlling the transistor **110**. CLOCK may vary from a first frequency f**0** to a second frequency f**1** with changes to the sensed voltage. Responsive to the comparison result having the value of logic **1** and the clock signal provided by the clock generator **206** having a value of logic **1**, the AND logic circuit **208** provides a logic result having a value of logic **1**. Responsive to either the comparison result having a value of logic **0** or the clock signal provided by the clock generator **206** having a value of logic **0**, the AND logic circuit **208** provides a logic result having a value of logic **0**. Responsive to the logic result having the value of logic **1**, the one-shot circuit **210** provides a signal pulse to the gate driver **202**. The signal pulse may have a pulse width specified according to the on-time value received by the one-shot circuit **210**.

[0032] Responsive to the signal pulse provided by the one-shot circuit **210** having an asserted value, the gate driver **202** provides the control signal to the transistor **110** to cause the transistor **110** to be conductive. The transistor **110** being conductive creates a short circuit event from the capacitor **130**, through the transistor **106** and the transistor **110** to ground (e.g., the ground terminal of the power source **102**) to partially discharge the capacitor **130**. The above operation may repeat until the sensed voltage decreases to approximately equal, or be less than, the stop voltage select value.

[0033] As described above, in some examples, the drive circuit **200** includes fault detection, implemented at least in part by the comparator **212**. For example, the reference signal is summed with a fault offset value to form a fault reference signal. While FIG. **2** includes a discrete summation component to add the reference signal and the fault offset value, the summation may instead be inherent, implemented by coupling two analog signal lines together. The comparator **212** receives the sensed voltage and the fault reference signal. Responsive to the sensed voltage having a value greater than the fault reference signal, the comparator **205** provides a comparison result having a value of logic **1**. Responsive to the sensed voltage not having a value greater than the fault reference signal, the comparator **205** provides the comparison result have a value of logic **0**. Responsive to the fault signal having the value of logic **0**, the drive circuit **200** determines that a fault condition related to discharge of the capacitor **130** does not exist. Responsive to the fault signal having the value of logic **1**, the drive circuit **200** determines that a fault condition related to discharge of the capacitor **130** exists. Responsive to determining that the fault condition exists, the drive circuit **200** may cease discharging, or attempting to discharge, the capacitor **130**. In various examples, the drive circuit **200** may cease discharging, or attempting to discharge, the capacitor **130** responsive to other events, such as determining that a temperature of the transistor **110**, or any other component of the drive circuit **200** or system **100** exceeds a threshold value.

[0034] FIG. **3** is a diagram **300** of signal waveforms, in accordance with various examples. In an example, the signals of the diagram **300** may be provided in a system such as the system **100**. In an example, the diagram **300** includes a signal V.sub.GS, representative of a gate-to-source voltage of the transistor **110**, a signal I.sub.DRAIN, representative of a drain current of the transistor **110**, a signal V.sub.DC_LINK, representative of a voltage of the capacitor **130**, and a signal E.sub.DIS, representative of an amount of energy discharged from the capacitor **130**. As shown by the diagram **300**, responsive to an increase in V.sub.GS and I.sub.DRAIN, V.sub.DC_LINK decreases in value by an amount ΔV and E.sub.DIS increases. As further shown by the diagram **300**, an on-time of the transistor **110**, based on the pulsed signal provided by the one-shot circuit **210**, is t.sub.on. In an example, a maximum voltage of the control signal provided to the transistor **110** may be selected to control a peak drain current (I.sub.PEAK) of the transistor **110**, such as to control, along with t.sub.on, an amount of energy discharged (EPP) from the capacitor **130** for each signal pulse provided by the one-shot circuit **210**.

[0035] FIG. **4** is a diagram **400** of signal waveforms, in accordance with various examples. In an example, the signals of the diagram **400** may be provided in a system such as the system **100**. In an example, the diagram **400** includes a signal AD_EN, representative of the active signal discharge enable signal, a signal V.sub.LDO, representative of a maximum value for the control signal, a signal envelope of V.sub.GS, a signal representative of a signal envelope of a frequency of the control signal (Freq), a signal envelope of I.sub.DRAIN, V.sub.DC_LINK, and E.sub.DIS. As shown by the diagram **400**, responsive to assertion of AD_EN, during a setup time t.sub.AD_SU, V.sub.LDO decreases in value, such as to limit I.sub.PEAK of the transistor **110**, as described above. At a completion of the setup time, the drive circuit **200** controls the transistor **110** according to PFM, as described above, to discharge the capacitor **130**. As the drive circuit **200** controls the transistor **110** to discharge the capacitor **130**, V.sub.GS and I.sub.DRAIN increase in value, as shown by the signal envelopes of V.sub.GS and I.sub.DRAIN. As shown by the signal envelope of Freq, the pulse frequency of the control signal by which the drive circuit **200** controls the transistor **110** dynamically increases to provide an approximately constant-power discharge, as shown by the approximately linear E.sub.DIS as V.sub.DC_LINK decreases in value. As further shown by FIG. **4**, the active discharge of the capacitor **130** may continue for an active discharge time (t.sub.AD) until V.sub.DC_LINK has decreased to approximately equal the stop voltage select value. The capacitor **130** may continue to discharge passively following an end of t.sub.AD.

[0036] FIG. **5** is a diagram **500** of signal waveforms, in accordance with various examples. In an example, the signals of the diagram **500** may be provided in a system such as the system **100**. In an example, the diagram **500** includes the signals of the diagram **400**, as well as the fault reference signal V.sub.REF, as described above herein. As shown by the diagram **500**, responsive to assertion of AD_EN, during a setup time t.sub.AD_SU, V.sub.LDO decreases in value, such as to limit I.sub.PEAK of the transistor **110**, as described above. At a completion of the setup time, the drive circuit **200** controls the transistor **110** according to PFM, as described above, to discharge the capacitor **130**. As the drive circuit **200** controls the transistor **110** to discharge the capacitor **130**, V.sub.GS and I.sub.DRAIN increase in value, as shown by the signal envelopes of V.sub.GS and I.sub.DRAIN. As shown by the signal envelope of Freq, the pulse frequency of the control signal by which the drive circuit **200** controls the transistor **110** dynamically increases to provide an approximately constant-power discharge. However, at time t**1**, a difference in value between VSENSE and V.sub.REF exceeds the offset value, as described above. Responsive to the difference in value between VSENSE and the V.sub.REF exceeding the offset value (shown in FIG. **5** as FAULT OFFSET LEVEL), a fault signal (FLT) is asserted, or provided having a value of logic **0**. Responsive to assertion of the fault signal, active discharge of the capacitor **130** by the low-side drive circuit **108** (or more generally, a drive circuit **200**, as described above herein) may be disabled. In some examples, the drive circuit for which active discharge is disabled may reattempt active discharge after a programmed amount of time, or another drive circuit (e.g., one, or both of, the low-side drive circuits **116**, **124** when active discharge is disabled for the low-side drive circuit **108**) may attempt active discharge, of the capacitor **130**.

[0037] FIG. **6** is a flowchart of a method **600**, in accordance with various examples. In an example, the method **600** is representative of at least some operations performed by a drive circuit, such as the drive circuit **200**. For example, the method **600** may be representative of at least some operations performed by the drive circuit **200** implemented as a low-side drive circuit **108**, **116**, **124**, as described above herein. In at least some examples, the method **600** is implemented to discharge a capacitor, such as the capacitor **130**, via a series of short-circuit events. In some examples, the discharge is performed according to PFM to provide an approximately constant, or linear, rate of power discharge.

[0038] At operation **602**, a sensed voltage is received. In some examples, the sensed voltage is received from a voltage sensing circuit, such as a voltage divider. The sensed voltage may be representative of, proportional to, or otherwise indicate a value of a voltage of a capacitor, such as a DC link capacitor.

[0039] At operation **604**, the sensed voltage is compared to a reference voltage to provide a comparison result. In some examples, the reference voltage is a reference signal waveform indicating a discharge curve for discharging the capacitor. In some examples, the reference voltage may be determined at a particular point in time based on a discharge time select value and a stop voltage select value, which may be obtained from any suitable source.

[0040] At operation **606**, responsive to the comparison result having an asserted value and a clock signal having an asserted value, a pulse signal is provided. In some examples, the clock signal is provided based on a clock frequency limit value that may limit a frequency of the pulse signal. The pulse signal may be provided, for example, by a one-shot circuit having a constant on-time.

[0041] At operation **608**, a transistor is controlled, according to the pulse signal, to be conductive for the on-time to partially discharge the capacitor. In some examples, controlling the transistor to become conductive creates a short-circuit event during which energy discharges from the capacitor, decreasing a voltage of the capacitor (and, consequently, decreasing the sensed voltage).

[0042] In this description, the term "couple" may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0043] A device that is "configured to" perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0044] A circuit or device that is described herein as including certain components may instead be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0045] While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0046] Uses of the phrase "ground voltage potential" in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, "about," "approximately" or "substantially"

preceding a parameter means being within +/−10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.
[0047] As used herein, the terms “terminal,” “node,” “interconnection,” “pin,” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device, or a semiconductor component. Furthermore, a voltage rail or more simply a “rail,” may also be referred to as a voltage terminal and may generally mean a common node or set of coupled nodes in a circuit at the same potential.

Claims

1. A circuit, comprising: a reference signal generator having first and second input terminals and an output terminal, the first input terminal of the reference signal generator configured to receive a discharge time select value, and the second input terminal of the reference signal generator configured to receive a stop voltage select value; a comparator having first and second input terminals, and an output terminal, the first input terminal of the comparator configured to receive a sensed voltage, and the second input terminal of the comparator coupled to the output terminal of the reference signal generator; a one-shot circuit having first and second input terminals, and an output terminal, the first input terminal coupled to the output terminal of the comparator, and the second input terminal configured to receive an on-time value; and a gate driver having an input terminal and an output terminal, the input terminal of the gate driver coupled to the output terminal of the one-shot circuit.
2. The circuit of claim 1, further comprising: an AND logic circuit having first and second input terminals, and an output terminal, the first input terminal of the AND logic circuit coupled to the output terminal of the comparator, and the output terminal of the AND logic circuit coupled to the first input terminal of the one-shot circuit; and a clock generator having an output terminal coupled to the second input terminal of the AND logic circuit.
3. The circuit of claim 1, further comprising a second comparator having first and second inputs, and an output terminal, the first input terminal of the second comparator coupled to the first input terminal of the comparator, the second input terminal configured to receive a fault reference signal formed by adding a fault offset level value to a reference signal provided by the reference signal generator, and the output terminal of the second comparator coupled to a second input terminal of the gate driver.
4. The circuit of claim 1, further comprising a voltage sense circuit having first and second terminals, and an output terminal, the second terminal of the voltage sense circuit coupled to a ground terminal, and the output terminal of the voltage sense circuit coupled to the first input terminal of the comparator.
5. The circuit of claim 4, wherein the voltage sense circuit comprises a voltage divider having an output coupled to the first input terminal of the comparator.
6. The circuit of claim 4, further comprising a capacitor having first and second terminals, the first terminal of the capacitor coupled to the first terminal of the voltage sense circuit, and the second terminal of the capacitor coupled to the ground terminal.
7. The circuit of claim 4, further comprising a transistor having first and second terminals, and having a control terminal, the first terminal of the transistor coupled to the first terminal of the voltage sense circuit, the second terminal of the transistor coupled to the ground terminal, and the control terminal of the transistor coupled to the output terminal of the gate driver.
8. A circuit, configured to: receive a sensed voltage representative of a voltage of an energy storage circuit; compare the sensed voltage to a reference signal representative of a discharge profile for the energy storage circuit; and responsive to the sensed voltage having a value greater than the reference signal, control a gate driver to provide a control signal configured to discharge the energy

storage circuit for a constant on-time period determined to cause a power discharge rate of the energy storage circuit to remain constant.

9. The circuit of claim 8, wherein the control signal is a pulse frequency modulation (PFM) signal having a frequency determined to cause the power discharge rate of the energy storage circuit to remain constant.

10. The circuit of claim 8, wherein the circuit is configured to control the gate driver to provide the control signal responsive to the sensed voltage having a value greater than the reference signal and responsive to a clock signal having an asserted value, wherein the clock signal limits a frequency of the control signal.

11. The circuit of claim 8, wherein the circuit is configured to compare the sensed voltage to a fault reference signal and, responsive to the sensed voltage exceeding the fault reference signal, provide a fault signal indicating a fault has occurred related to discharge of the energy storage circuit.

12. The circuit of claim 8, wherein the circuit is configured to provide the control signal at a control terminal of a transistor to cause the transistor to become conductive, creating a short circuit between the energy storage circuit and a ground terminal.

13. The circuit of claim 12, wherein the energy storage circuit is a direct current (DC) link capacitor of an electric vehicle power system.

14. The circuit of claim 8, wherein the circuit is configured to determine the discharge profile for the energy storage circuit according to a discharge time select value and a stop voltage select value to cause the energy storage circuit to be discharges to a voltage less than or equal to the stop voltage select value within a time period specified by the discharge time select value.

15. A system, comprising: a switch having first and second terminals, the second terminal of the switch coupled to a positive power terminal; a capacitor having first and second terminals, the first terminal of the capacitor coupled to the first terminal of the switch, and the second terminal of the capacitor coupled a ground terminal; a voltage sense circuit having first and second terminals and an output terminal, the first terminal of the voltage sense circuit coupled to the first terminal of the switch, and the second terminal of the voltage sense circuit coupled to the ground terminal; a high-side drive circuit having an input terminal and an output terminal; a first transistor having first and second terminals, and a control terminal, the first terminal of the first transistor coupled to the first terminal of the switch, and the control terminal of the first transistor coupled to the output terminal of the high-side drive circuit; a low-side drive circuit having first and second input terminals and an output terminal, the first input terminal of the low-side drive circuit coupled to the output terminal of the voltage sense circuit; and a second transistor having first and second terminals, and a control terminal, the first terminal of the second transistor coupled to the second terminal of the first transistor, the second terminal of the second transistor coupled to the ground terminal, and the control terminal of the second transistor coupled to the output terminal of the low-side drive circuit.

16. The system of claim 15, wherein, responsive to receiving a safe state enable signal having an asserted value at the input terminal of the high-side drive circuit, the high-side drive circuit is configured to control the first transistor to remain in a conductive state.

17. The system of claim 15, wherein, responsive to, receiving an active discharge enable signal having an asserted value at the second input terminal of the low-side drive circuit, the low-side drive circuit is configured to modulate a control signal provided at the control terminal of the second transistor according to pulse frequency modulation (PFM) to discharge the capacitor.

18. The system of claim 17, wherein the low-side drive circuit is configured to discharge the capacitor according to a constant power discharge rate by creating a series of short circuit events between the capacitor and a ground voltage potential, the short circuit events having a constant on-time.

19. The system of claim 17, wherein the low-side drive circuit is configured to: receive a sensed voltage from the voltage sense circuit, the sensed voltage representative of a voltage of the capacitor; and responsive to the sensed voltage exceeding a value of a reference signal, provide a signal pulse to control the second transistor to be conductive for a constant on-time period.

20. The system of claim 17, wherein the low-side drive circuit is configured to: receive a sensed voltage from the voltage sense circuit, the sensed voltage representative of a voltage of the capacitor; and responsive to the sensed voltage exceeding a value of a reference signal and a clock signal having an asserted value, provide a signal pulse to control the second transistor to be conductive for a constant on-time period.
